

# Piyush Chokker

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## EDUCATION

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K.R Mangalam University

BCA – AI & Data Science | CGPA: 7.0

K.R Mangalam University

MCA – AI & Machine Learning | CGPA: 7.2 (Average)

West Academy Sr. Sec. School

Class XII | 82%

## EXPERIENCE

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### Sundrop Brands | MIS Analyst – Automation

- Engineered MIS reporting automation using Python scripts and advanced Excel functions (VLOOKUP, Pivot Tables) for operational data, resulting in a 70%+ reduction in manual processing time.
- Conducted data validation and analysis on operational datasets to identify supply and sale bottlenecks and improve data integrity for procurement-related decision-making.
- Managed and integrated e-commerce sales/inventory data from major marketplace APIs (Amazon, Flipkart, etc.), creating centralized datasets critical for forecasting and raw material procurement planning.

Tools: Excel, Python, Power BI, Google Forms

## PROJECTS

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### [AskOllie](#) | [Production Application](#) | [Paid Internship](#)

- Built a production-ready AI-powered RAG chatbot using Next.js 16, FastAPI, Supabase Auth, Redis, Qdrant Vector DB, and SSE streaming, supporting around 8,000 monthly users.
- Developed v1 of the system using LangChain, LangGraph, and LlamaIndex for orchestration and retrieval; iteratively replaced most framework-dependent components with custom-built implementations to improve performance, control, and reduce overhead.
- Implemented secure JWT authentication, protected chat APIs, user-specific session validation, and access-controlled document retrieval.
- Engineered a metadata-filtered Qdrant vector retrieval pipeline, achieving sub-60 ms document chunk retrieval and improving relevance for context-aware RAG answer generation.
- Implemented a 3-layer caching architecture covering session caching, semantic caching, and backend/runtime cache optimization to reduce repeated computation.
- Optimized chat performance from ~5–10 seconds to 700–1000 ms total API response time, with 200–700 ms first-token streaming latency.
- Evaluated RAG quality using RAGAS, achieving approximately 0.92 faithfulness, 0.89 answer relevancy, 0.86 context precision, and 0.84 context recall.
- Delivered the project end-to-end: system design, frontend development, backend API development, authentication, vector database integration, caching, deployment, monitoring, debugging, latency optimization, and RAG quality evaluation.

### [PhantomPointer](#)

- Developed an Invisible Mouse system using computer vision to replace a physical mouse by tracking hand movements on a flat surface.
- Implemented real-time gesture recognition and motion tracking with OpenCV to simulate cursor movement and click actions.
- Designed an intuitive, low-latency interaction model enabling users to control the system seamlessly without traditional hardware.

### Web Scraping & Data Analysis

- Performed Exploratory Data Analysis (EDA) and built ML models on the US Accidents dataset to uncover patterns, trends, and key factors affecting road incidents.

- Developed web scraping pipelines using BeautifulSoup and Selenium to extract and analyze US stock market data and Indian options trading data.
- Applied data preprocessing, feature engineering, and visualization techniques to derive actionable insights and support predictive modeling.

### **Model Deployment & Fine-Tuning on NVIDIA DGX Spark**

- Set up an NVIDIA DGX Spark AI infrastructure environment with Triton Inference Server, GPU runtime configuration, model-serving dependencies, and deployment tooling.
- Deployed a quantized AI model on DGX Spark to improve inference efficiency, reduce memory usage, and support faster model execution using vLLM.
- Fine-tuned a small language model on DGX Spark, covering dataset preparation, training configuration, GPU utilization, and model validation.
- Implemented an optimized model inference workflow using Triton for serving, testing, and evaluating model responses in a GPU-accelerated environment.
- Gained hands-on experience with AI infrastructure setup, GPU-based inference, quantization, model fine-tuning, and production-oriented model serving workflows.

## **SKILLS & ABILITIES**

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**Programming Languages:** Python (NumPy, Pandas, Matplotlib, Seaborn), SQL

**Machine Learning & Deep Learning:** Scikit-Learn, TensorFlow, PyTorch

**Data Analysis & Visualization:** Matplotlib, Seaborn, Power BI, Tableau

**Big Data & Cloud Technologies:** Hadoop, Cloudera

**GenAI & AI:** OpenCV, LangChain, Streamlit, Ollama, Hugging Face

**Web Scraping & Automation:** BeautifulSoup, Selenium

**Version Control & DevOps:** Git, GitHub (CI/CD), Docker

**Deployment & Backend:** FastAPI, REST APIs, Railway, Vercel, Render, AWS

## **ACHIEVEMENTS**

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- Secured 2nd place out of 150 participants in the University Tech Quiz Competition, demonstrating strong knowledge of computer science.
- Designed and constructed a compact sound amplifier that won 1st position at the School Tech Exhibition.
- Participated in Mind Menders with Invisible Table Mouse project.

## **COLLABORATIVE TOOLS**

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Git, GitHub, Jupyter Notebook, Anaconda, Google Colab, Google Sheets, MS Excel, MS Word, PowerPoint, Outlook